REPUBLIC POLYTECHNIC

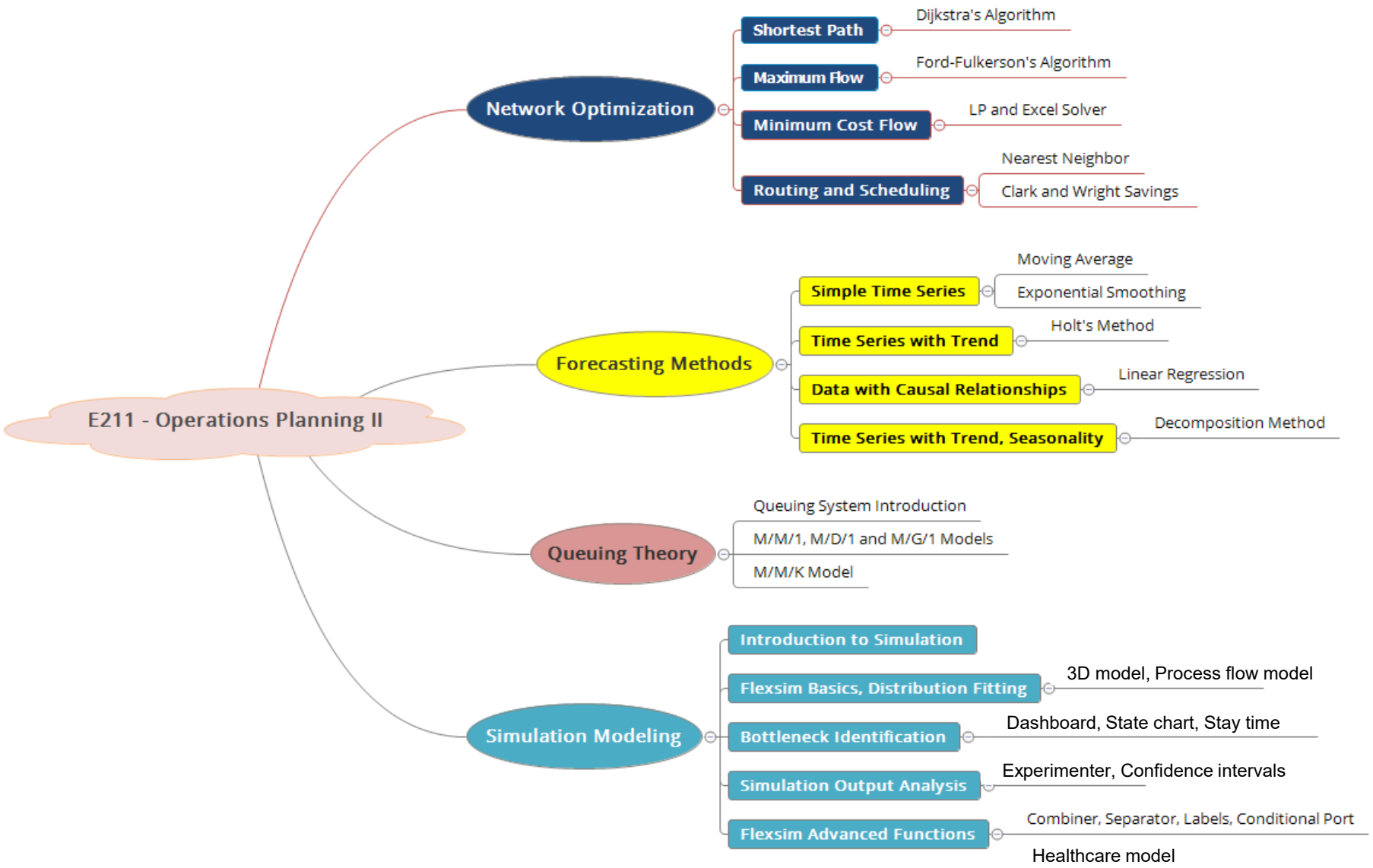
Lesson 8

Modeling & Simulation of Bottling Line (II)

E211 – Operations Planning II

REPUBLIC POLYTECHNIC

E211 Topic Tree



Recap on Bottling Line Model

REPUBLIC POLYTECHNIC

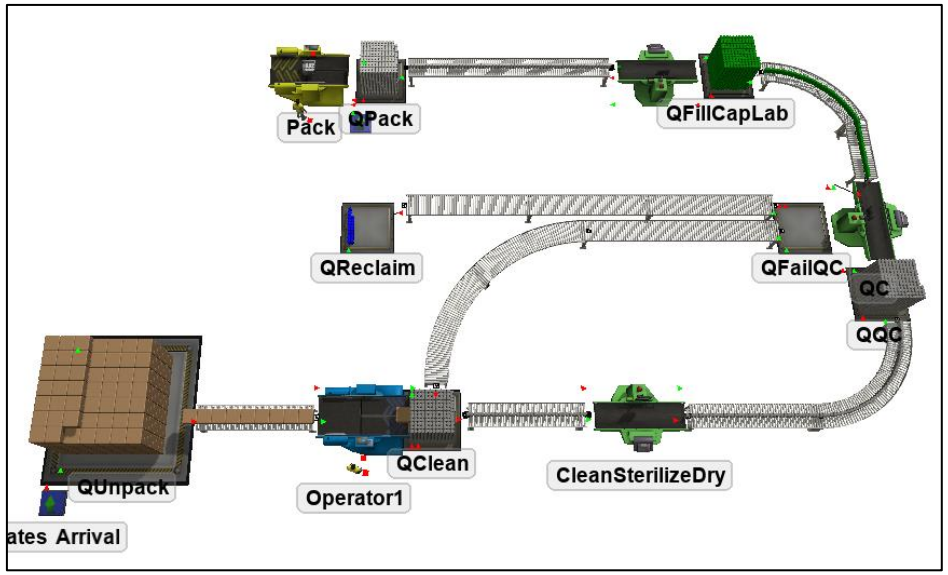
Model Building Steps

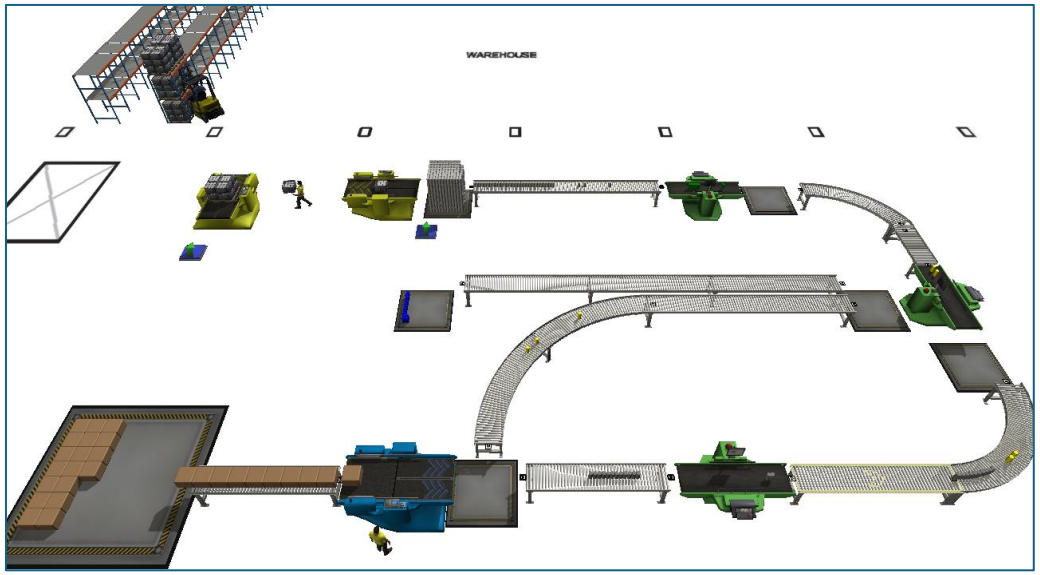
1. Import model background
2. Create item arrivals
3. Use Separator to model unpacking
4. Connect production flow with conveyors
5. Set Processor parameters
6. Create and use item QC label
7. Set item color with QC label
8. Use Combiner object to model packing
9. Use Combiner object to model palletization
10. Create storage Racks
11. Model Task Executers
12. Create Dashboard
13. Set up and run Experimenter

Covered in Lesson 7

Today's coverage

Remaining: Palletization & Putaway

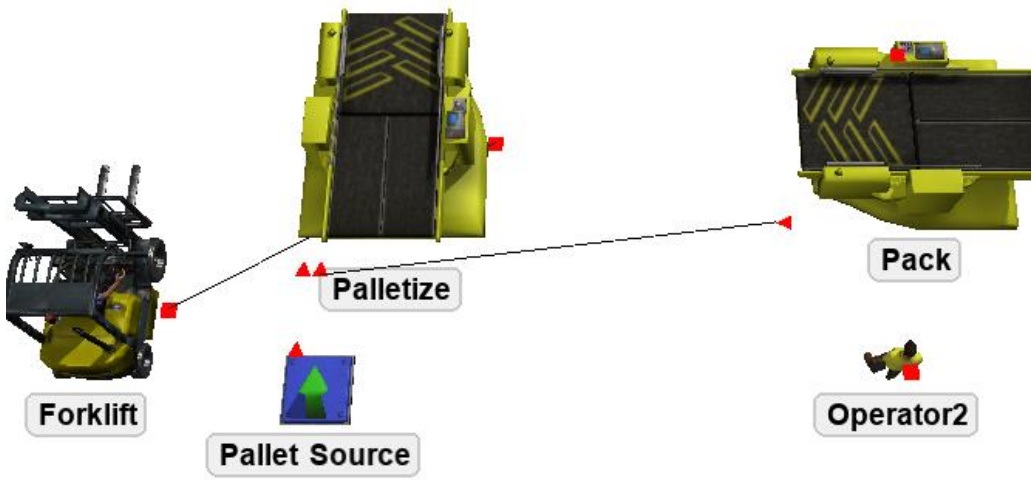




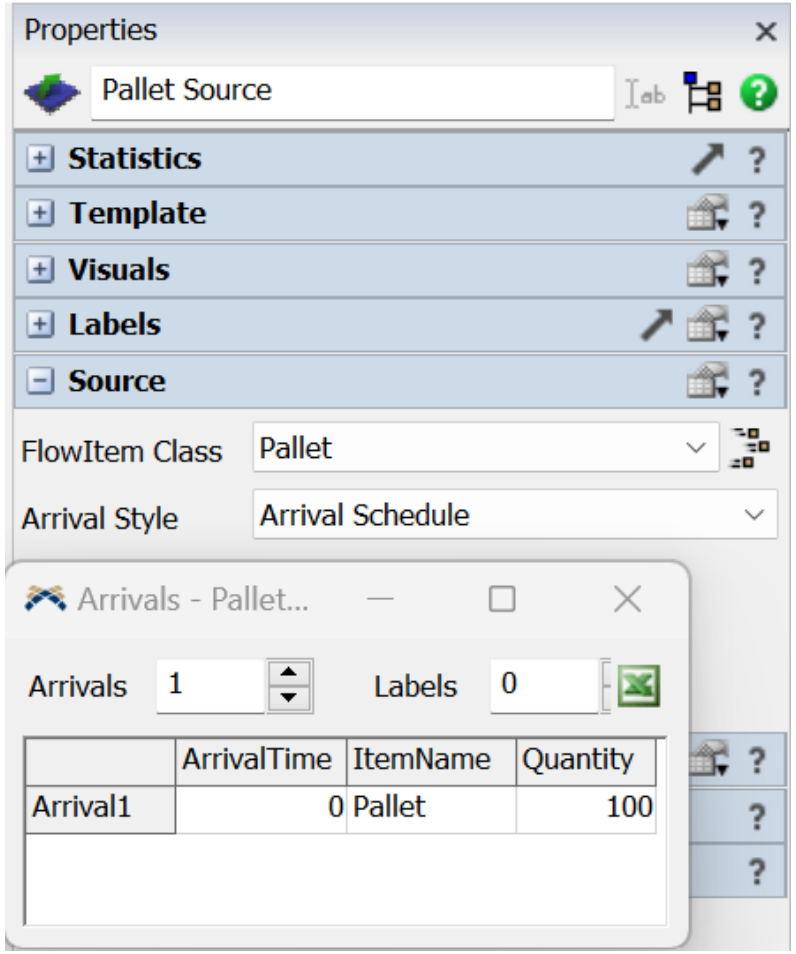
Palletization & Putaway

9. Palletization of Bottle Crates

1. Create a new Source and named it ‘Pallet Source’.
2. Create a new Combiner and named it ‘Palletize’.
3. Connect Pallet Source to Palletize first. Then, connect Pack to Palletize.
4. Create a Transporter and rename as ‘Forklift’.
5. Connect Forklift to Palletize via center port connection.



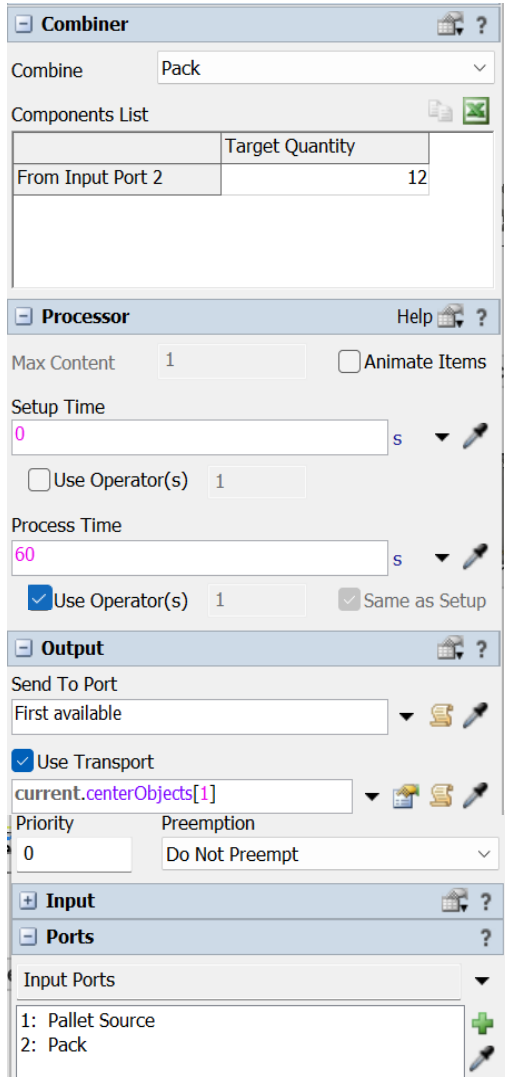
Pallet Source Settings



Pallet Source:

1. Set FlowItem Class to ‘Pallet’
2. Set Arrival Style to ‘Arrival Schedule’ and key in the data as shown

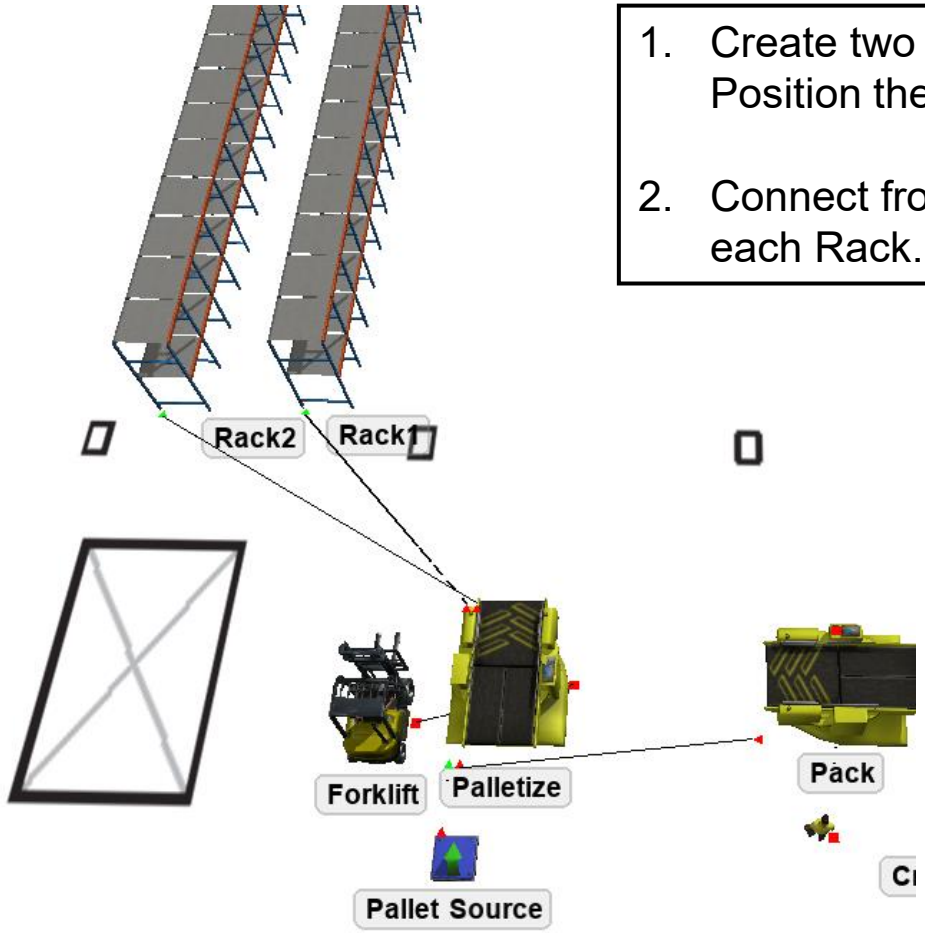
Palletize Settings



Palletize Combiner:

1. Set Combine mode to ‘Pack’. Enter ‘12’ in Target Quantity.
2. Key in Process Time = 60.
3. Check Use Operator (Forklift) under Process Time.
4. Check Use Transport under Send to Port.
5. Check that the Input Ports are correctly modeled.

10. Create Racks for Pallet Storage

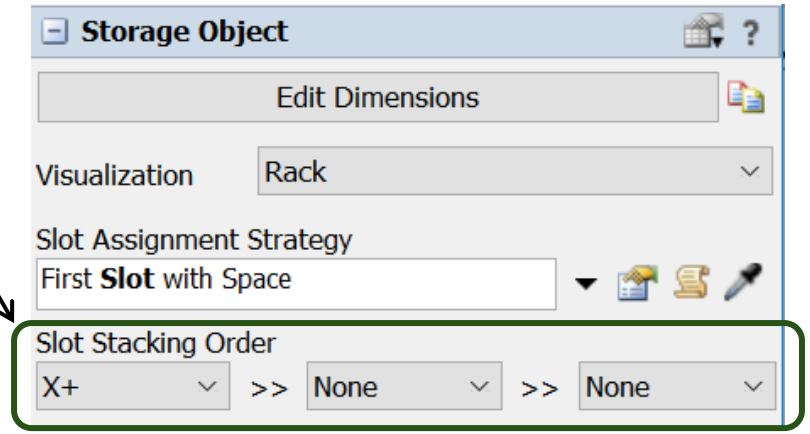


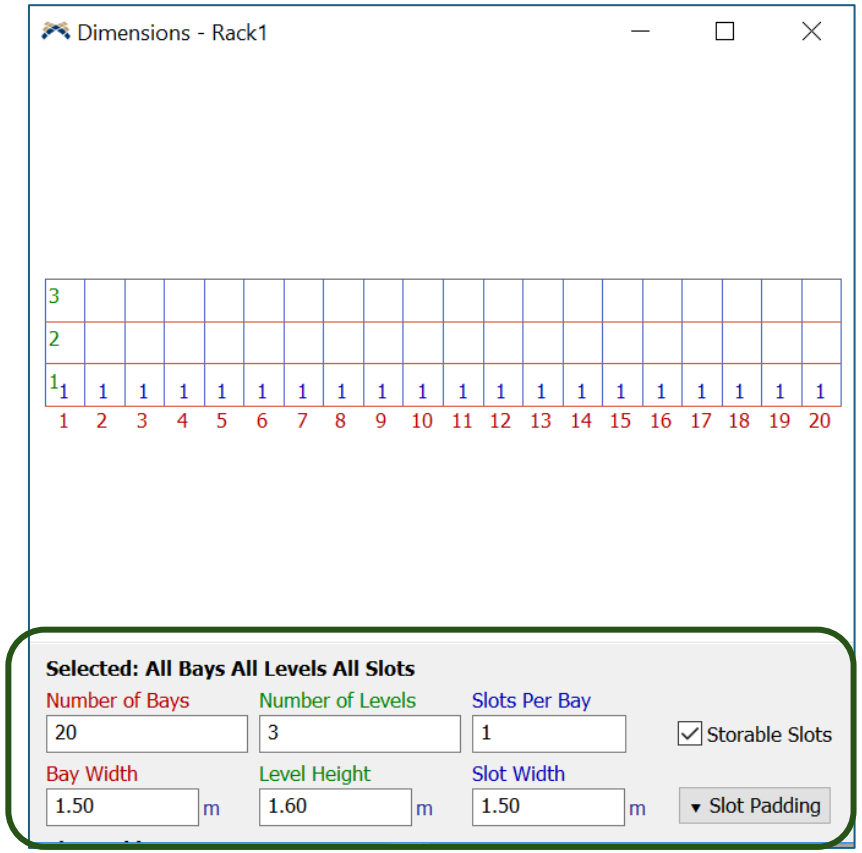
1. Create two Racks and Position them as shown.

2. Connect from Palletize to each Rack.

Customize Racks

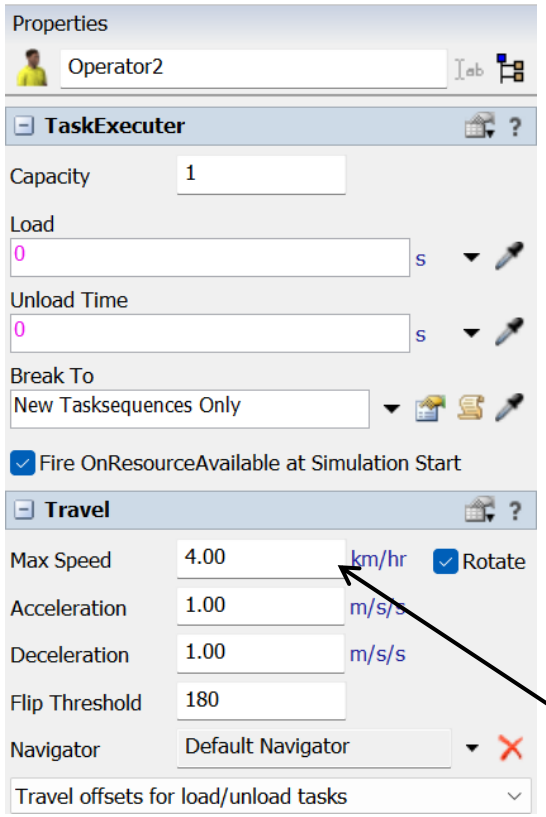
- In Rack properties, Storage Object, change the Slot Stacking Order as shown
- Click Edit Dimensions and key in the inputs as shown

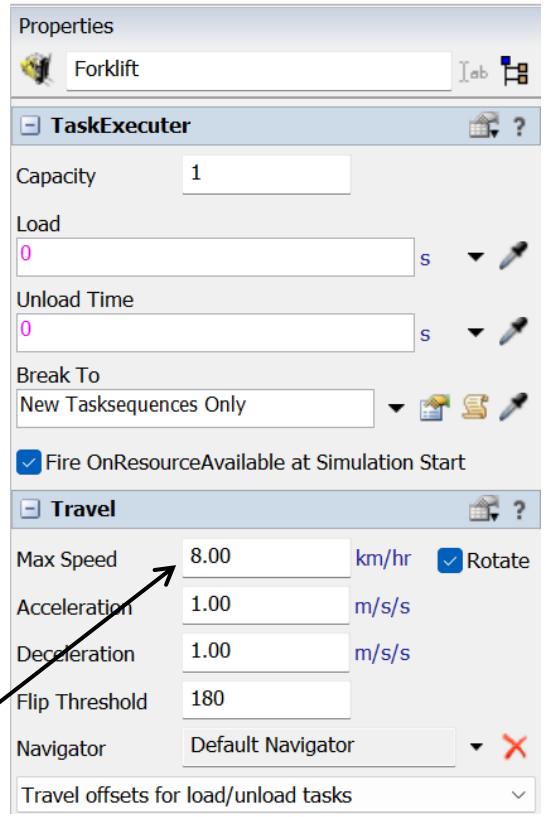




11. Modeling Task Executers

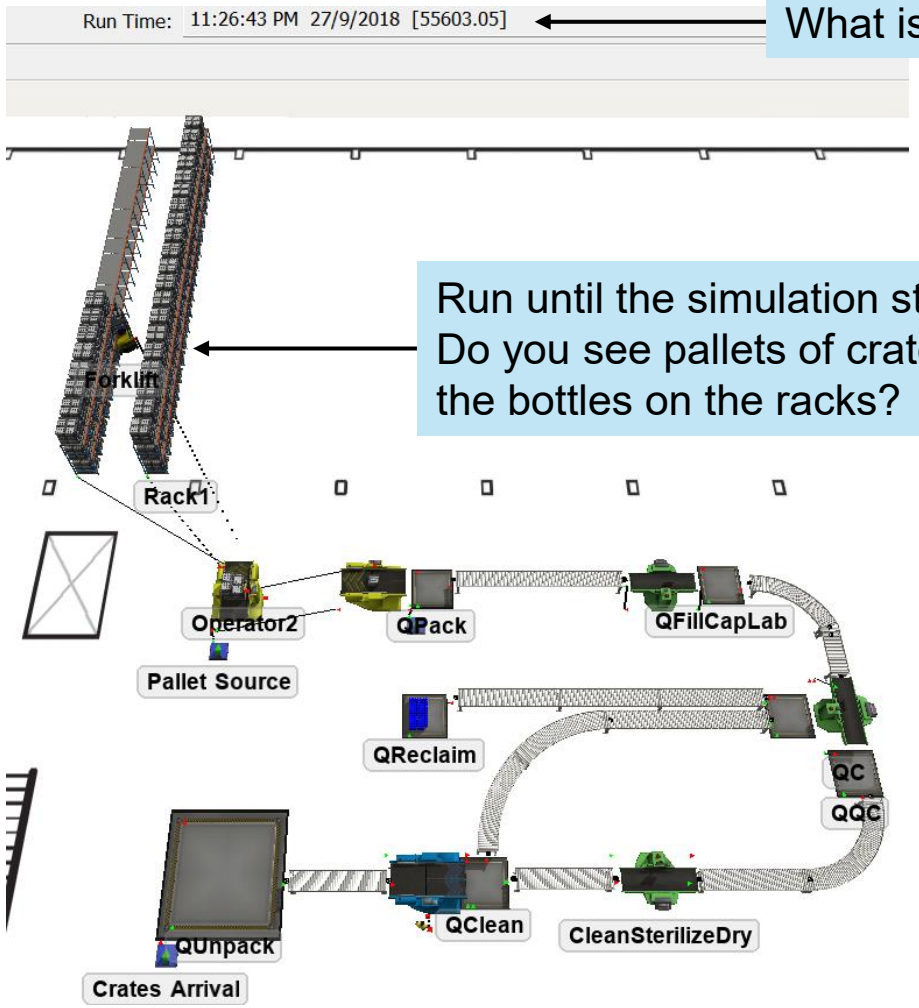
The capacity, speed, loading and unloading time of task executers (operators and transporters) should be modeled correctly.





Set Max speed of Operator and Transporter to 4km/hr & 8km/hr respectively

Model Checkpoint #3

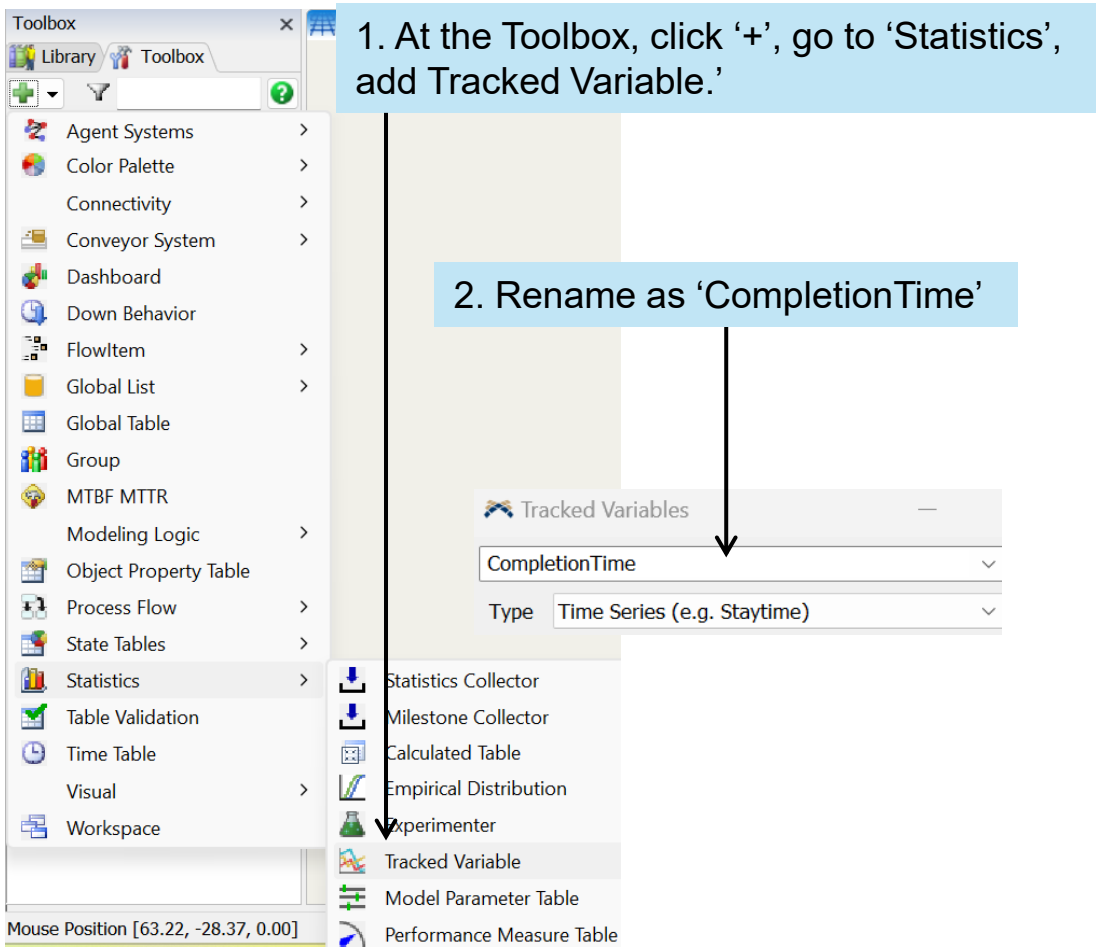


What is the stop time in your simulation?

Run until the simulation stops.
Do you see pallets of crates with the bottles on the racks?

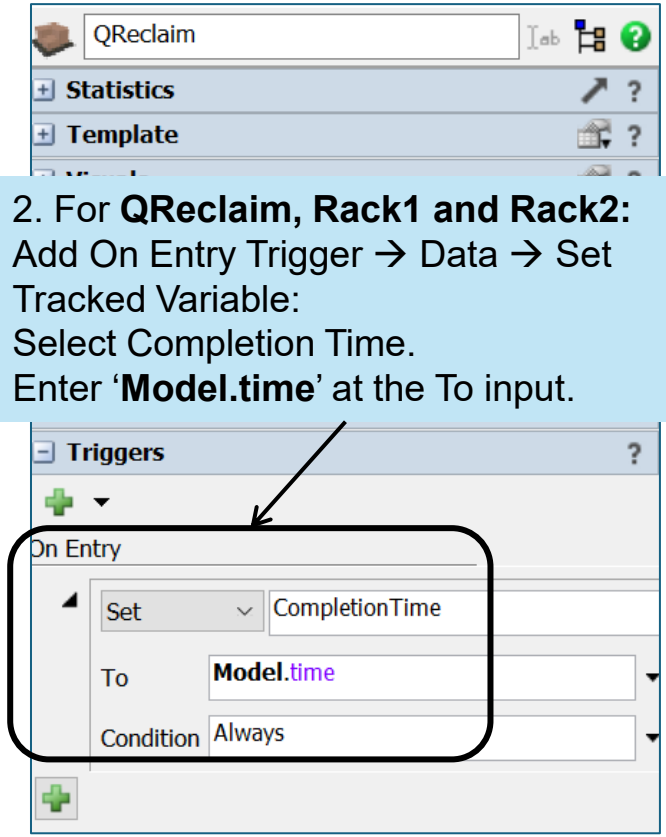
Create Dashboard

Create Completion Time Tracked Variable



1. At the Toolbox, click '+', go to 'Statistics', add Tracked Variable.'

2. Rename as 'CompletionTime'



2. For **QReclaim, Rack1 and Rack2:**
Add On Entry Trigger → Data → Set Tracked Variable:
Select Completion Time.
Enter '**Model.time**' at the To input.

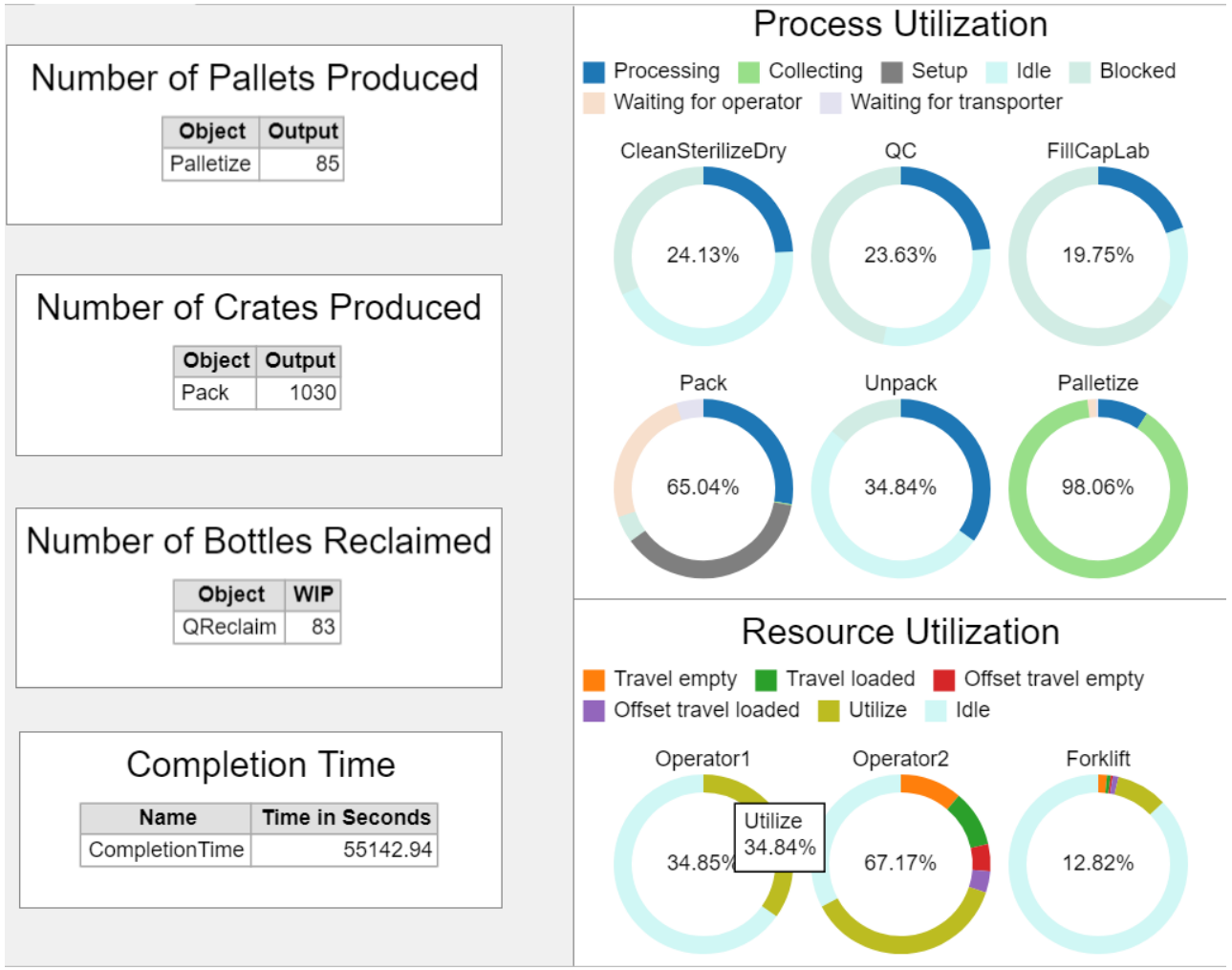
12. Create Dashboard

Set up the following statistics in the dashboard:

1. **Number of Pallets Produced** – Output Table (Palletize)
2. **Number of Crates Produced** – Output Table (Pack)
3. **Number of Reclaims** – Content Table (QReclaim)
4. **Completion Time** – Tracked Variable Table (Completion Time)
5. **Process Utilization** – State Pie Chart (all processors, combiners, separator)
6. **Resource Utilization** – State Pie Chart (all task executers)

Creating Charts in Dashboard

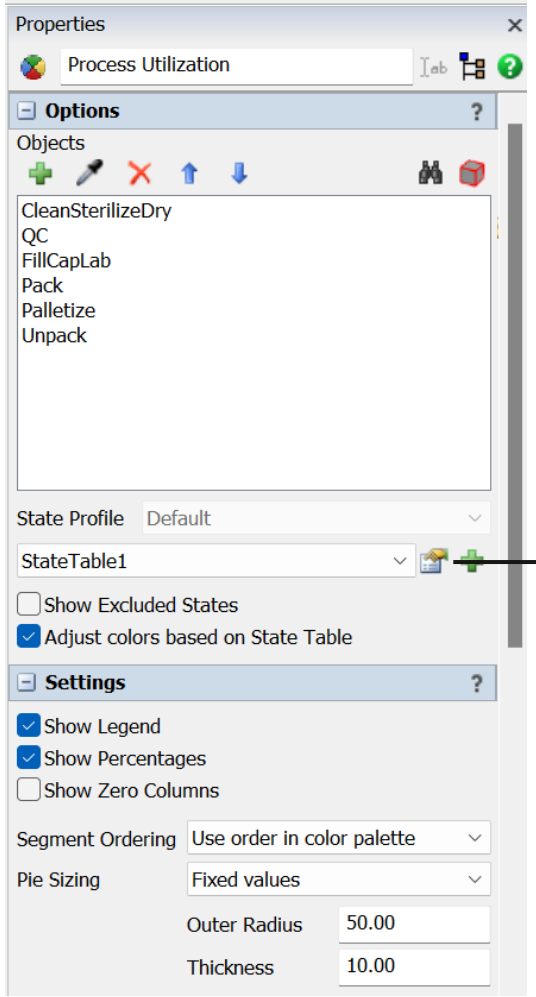
Video: https://tinyurl.com/2zpe3vbr



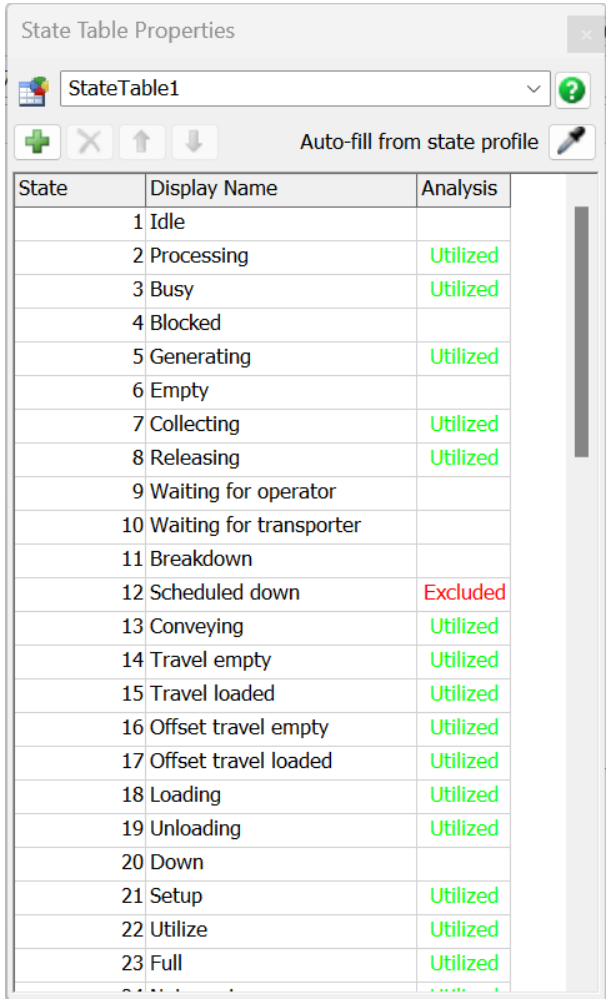
Percentage shown in the Pie Chart

= Total percentage utilization of all pre-defined 'Utilized' states

Object States



Click to open StateTable and select which states are to be defined as 'Utilized'

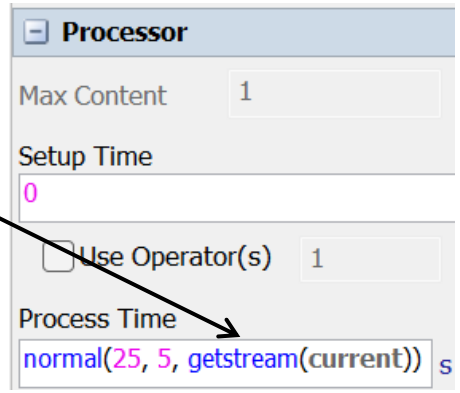


State	Display Name	Analysis
1	Idle	
2	Processing	Utilized
3	Busy	Utilized
4	Blocked	
5	Generating	Utilized
6	Empty	
7	Collecting	Utilized
8	Releasing	Utilized
9	Waiting for operator	
10	Waiting for transporter	
11	Breakdown	
12	Scheduled down	Excluded
13	Conveying	Utilized
14	Travel empty	Utilized
15	Travel loaded	Utilized
16	Offset travel empty	Utilized
17	Offset travel loaded	Utilized
18	Loading	Utilized
19	Unloading	Utilized
20	Down	
21	Setup	Utilized
22	Utilize	Utilized
23	Full	Utilized

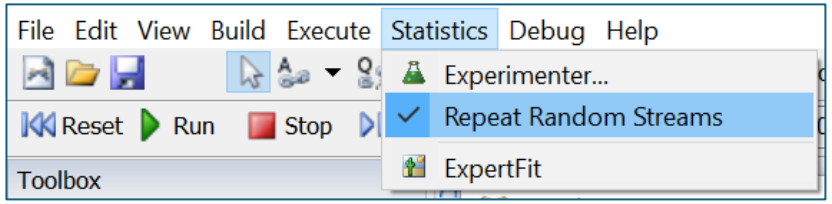
Experimenter

Randomness in Simulation

- In simulation, random number streams are used to generate random timings.



- If 'Repeat Random Streams' is checked, every time you run the simulation, you get the same result (outcome).



- Experimenter can be used to run multiple simulations with different possible outcomes.

13. Set Up Experimenter

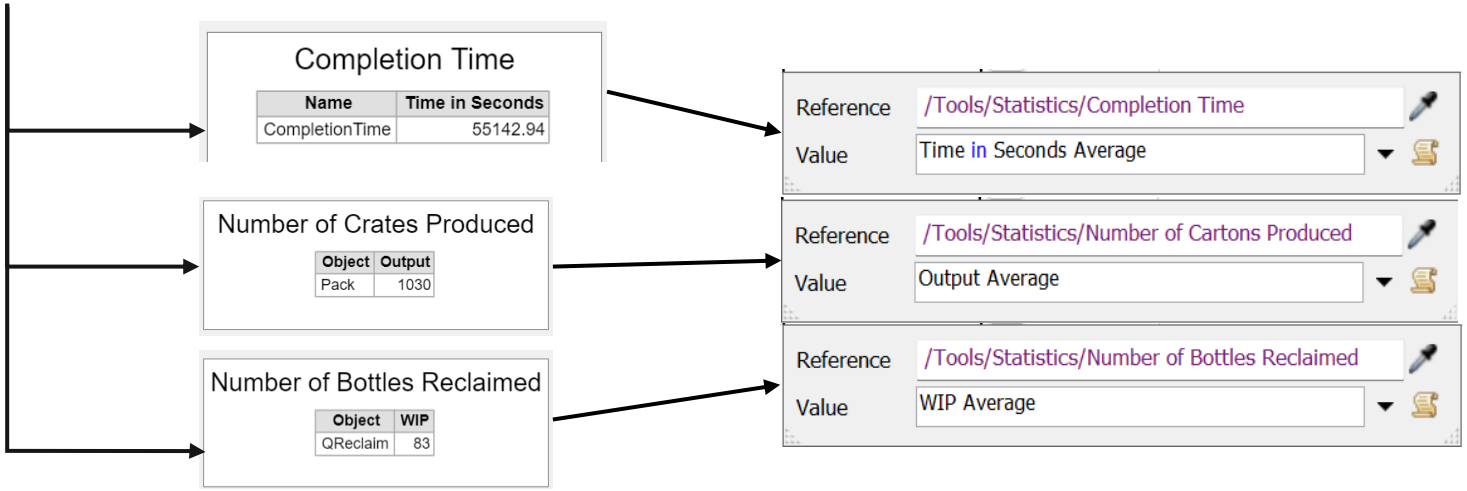
Before running the Experimenter, the Performance Measures must be defined. Go to Toolbox, double click on PerformanceMeasures and create:

1. Completion Time
2. Number of Crates Produced
3. Number of Bottles Reclaimed

Performance Measures 3

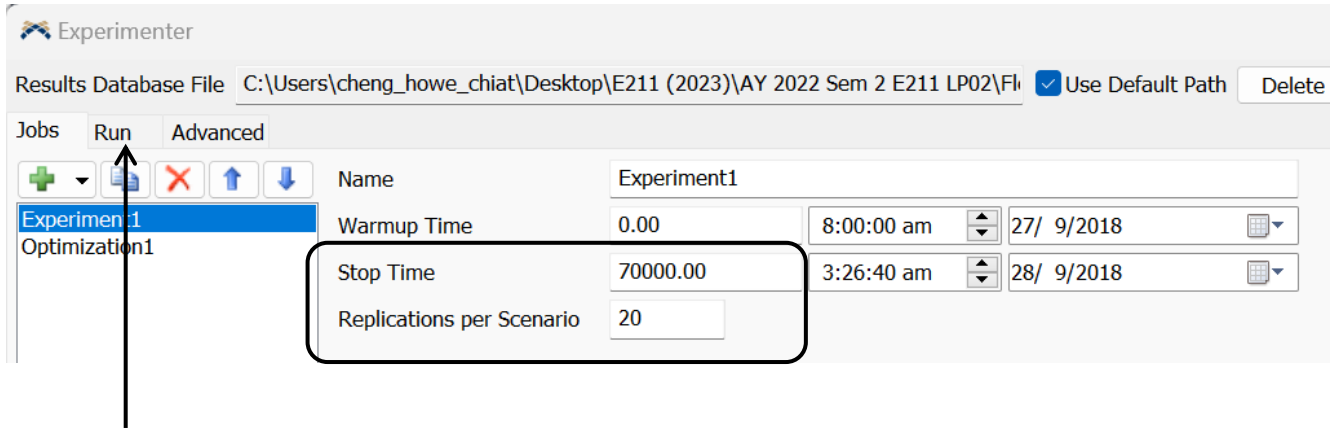
Name	Value	Display Units
Completion Time	0	
Number of Crates Produce	0	
Number of Bottles Reclaim	0	

For each Performance Measure, click on the Value cell, click the option button, and then use the Sampler Tool to select the relevant Dashboard chart.

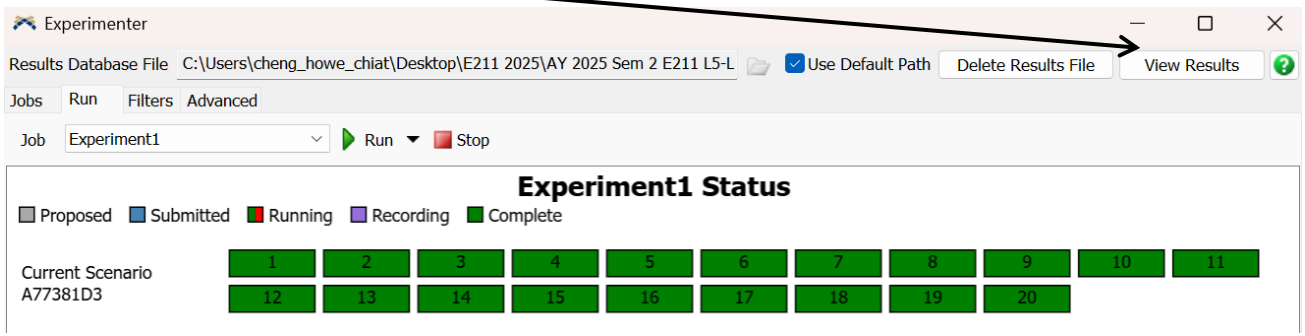


Experimenter Runs

- Go to Statistics on the menu bar, then click on Experimenter
- Key in number of replications = 20
- Key in a stop time value longer than the completion time (i.e. 70000)



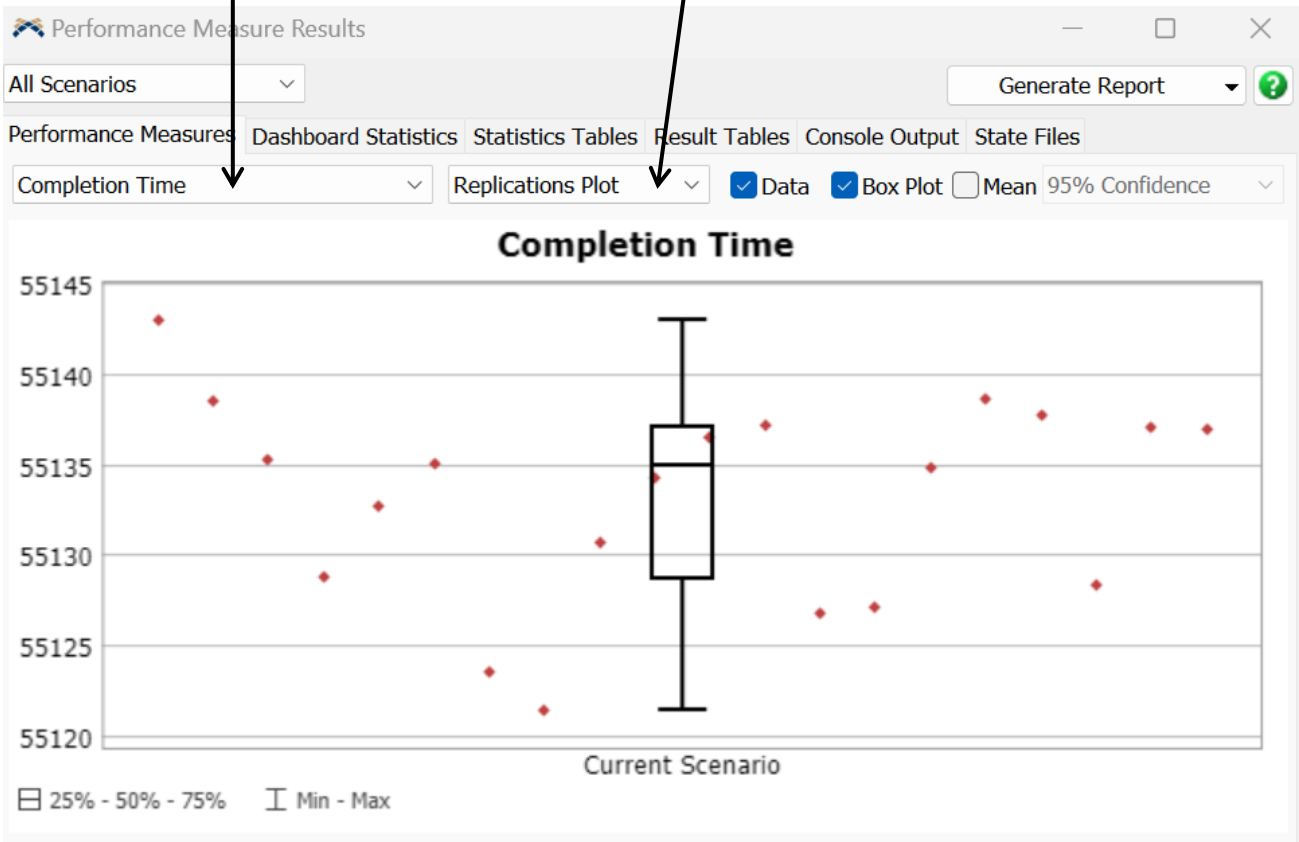
- Click on Run tab. Then click Run.
- Click View Results when all runs are completed



Experimenter Results

Click here to select Performance Measure

Click to select type of result



Data Summary Results

1. 95% confidence interval of Completion Time

Completion Time	Mean (95% Confidence Interval)	Sample Std Dev	Min	Max
Current Scenario	55133.23 ± 2.62	5.60	55121.51	55142.94

2. 95% confidence interval of Number of Crates Produced

Number of Crates Produced	Mean (95% Confidence Interval)	Sample Std Dev	Min	Max
Current Scenario	1029.550 ± 0.355	0.759	1028.000	1031.000

3. 95% confidence interval of Number of Bottles Reclaimed

Number of Bottles Reclaimed	Mean (95% Confidence Interval)	Sample Std Dev	Min	Max
Current Scenario	89.30 ± 3.06	6.53	76.00	101.00

Total number of bottles processed = 13 batches x 60 crates per batch x 12 bottles per crate
Therefore, average reclaim rate = 89.3/(60*13*12) *100% = 0.95%

Does the value make sense? 10% of bottles fail QC.

Experimentation

Help Justin study these additional proposed changes:

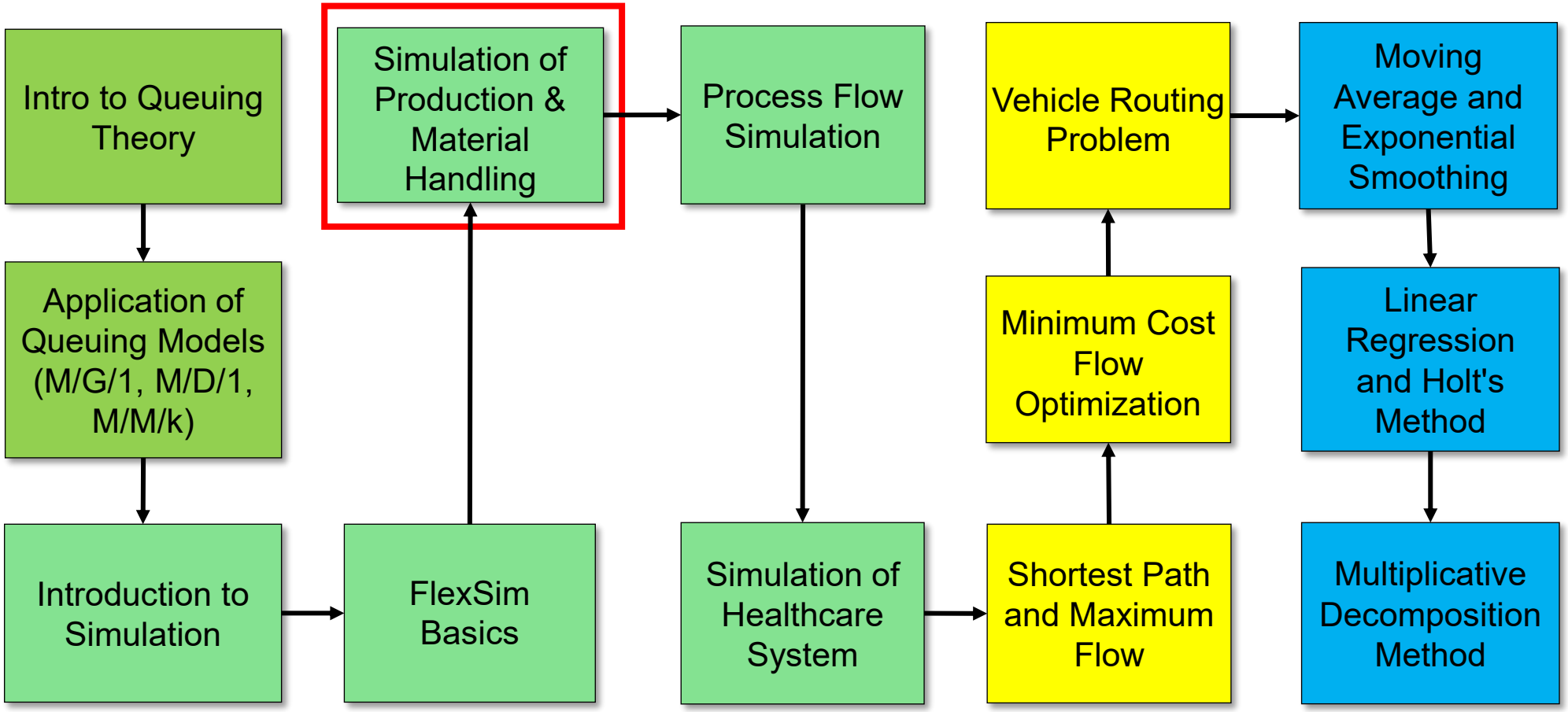
A. Install a conveyor to move items between Pack and Palletize. A Robot arm will perform the palletization. Pack will be fully automated (process time remains unchanged) so that there no need for setup. Operator2 is redeployed to perform unpacking at a newly setup station in parallel to the current unpack station.

B. To support reusables, the company wants to increase the use of recycled bottles. It is estimated that the QC pass rate will drop to 80% and more thorough cleaning for bottles is required. CleanSterlizeDry process time will follow Uniform Distribution (min = 20s, max = 30s).

C. To increase recycling rate, the maximum number of reworks allowed is increased to 2. For bottles that have failed QC 2 times, CleanSterlizeDry process time will be increased to a constant time of 45s (timing for 1st and 2nd cleaning follows the timing given previously). Assign the color black to bottles that fail 3 times (reclaimed bottles).

Learning Outcomes

At the end of the lesson, students should be able to:

- Construct a 3D production model in FlexSim to simulate material conveyance and processing.
- Apply modeling features in FlexSim to simulate differentiated flow, status tracking and appearance alteration.
- Perform multiple simulation runs to generate mean and confidence interval results.

E211 Progression